

Intelligent Soft Wearables

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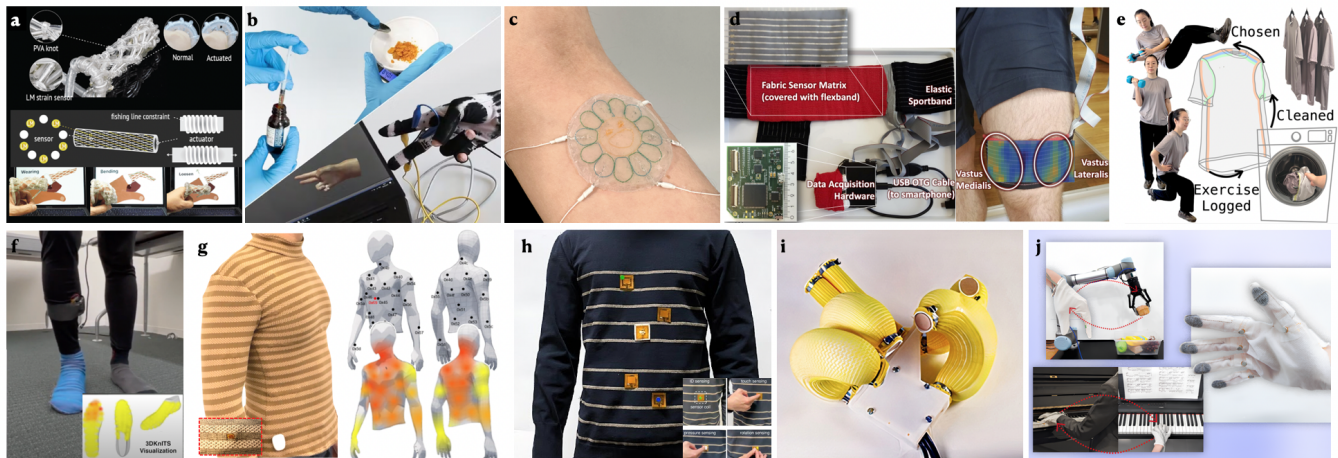


Figure 1: Showcasing the range of intelligent soft wearables spanning (a) recyclable electronics, (b) resistive fabrics, and (c) on-skin self-powered sensing; wearable platforms for (d) exercise monitoring and (e) loose-fitting sensing clothing; digital manufacturing of knit textiles and modular printed circuits for (f) tactile sensing, (g) physiological sensing, (h) on-body interactions, and (i) soft robotics; and finally, (j) an adaptive embroidered smart glove for haptic-enabled learning.

Abstract

Human bodies are almost always in contact with soft materials like clothing, for warmth, protection, self-expression, etc. Recent advancements in intelligent soft wearables have augmented these

on-body soft objects with computational functions and intelligence with little compromise on the softness and comfort of wearables, allowing prolonged wear. These innovations, which combine advanced soft sensor design, fabrication, and computational power, offer unprecedented opportunities to improve our health, productivity, and overall well-being with monitoring and assistive capabilities. However, the inherent physical properties of soft materials present unique challenges in achieving practical interactions. The complexity of intelligent soft wearables, multiplexing intricate designs, soft materials, flexible electronics, advanced signal processing algorithms, and machine learning models, necessitates collaborative



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efforts from experts across diverse domains. This workshop aims to bring together interested researchers and practitioners across relevant domains to discuss the challenges and opportunities of intelligent soft wearables.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing**.

Keywords

Wearables, Soft Technologies, Applied Machine Learning, e-Textiles, Personal Fabrication, Sensing, Actuation, Energy Harvesting.

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1 Introduction

The comfortable and conformable nature, essential factors in wearables' adoption or abandonment [1, 6], of soft wearables affords practical and prolonged uses of interactive interfaces [5, 18–20], behavior monitoring and learning systems [13, 21, 23, 26–29], assistive wearables [12, 15], human augmentation [11], and entertainment [7–9, 16], with great potential in HCI, healthcare, AR/VR, robotics, and space, among other applications. Powered by advanced machine learning techniques, these intelligent wearables, serving as sensing and actuation interfaces, unobtrusively extract information and generate optimal feedback.

Compared with rigid wearables (e.g., smartwatches and smart glasses), soft wearables (e.g., textiles, silicone rubber) offer the benefits of conformability and comfortable large area coverage, but pose additional obstacles. For example, soft sensors suffer from hysteresis, detachment, and permanent deformations due to wear and tear. Similarly, soft actuators face difficulties due to the need for robust design and close coupling with the substrate of interest (such as the skin) to provide effective feedback. Actuation in soft interfaces is particularly challenging as it requires a delicate balance between maintaining the desired softness for comfort and conformability while providing sufficient strength to deliver effective and reliable performance. Despite recent advances in material science, digital fabrication, electronics miniaturization, and applied machine learning that have enabled exciting intelligent soft wearable applications, critical research challenges (detailed in Section 3) still lie ahead for everyday adoptions of such systems by everyone. Addressing these human-centered challenges requires collaborative efforts from the design, scientific, and engineering communities as the design, fabrication, and application of intelligent soft wearables generally require domain-specific knowledge and cross-disciplinary integration [22].

At UIST 2024, we organized a successful workshop¹ on democratizing intelligent soft wearables [10] and attracted more than 40

attendees. Carrying forward the momentum, in this workshop, we would like to invite the UbiComp/ISWC community to merge expertise in a wide areas from technical HCI and design to technology acceptance. We aim to make this workshop serve as a platform to identify the challenges and opportunities for intelligent soft wearables research and foster collaboration.

2 Workshop Objectives

The goals and expected outcomes of our workshop include:

- Gathering researchers in relevant fields to share their findings, practices, and insights and showcase demos and Work in Progress (WIP);
- Discussing the grand challenges in designing, implementing, evaluating, and deploying intelligent soft wearables;
- Promoting intelligent soft wearable to the UbiComp/ISWC community;
- Fostering collaboration among researchers from various fields to enrich the research on intelligent wearable devices for HCI through diverse perspectives; and
- Planning for future events on related topics.

3 Workshop Topics: Opportunities and Challenges in Intelligent Soft Wearables

Topics of interest for this workshop include all aspects of soft intelligent wearables, from the design & implementation to applications and social acceptance & adoption. The workshop topics include (but are not limited to):

Trade-Offs between Functionalities and Softness/Wearability: Designing intelligent soft wearables involves navigating trade-offs between maximizing functionality and preserving softness and wearability.

Individual Differences → Generalization&Personalization: Humans have a wide range of body sizes and shapes, movement patterns, perceptions, physiological responses, behavior patterns, etc, making it difficult and unreasonable for universal generalization, given individual differences. Customization and/or calibration are often required to tailor personalized individual needs.

Closed-Loop Control with Sensors and Actuators: Close-loop control systems enable real-time dynamic interactions with the user: the sensors continuously monitor users' intent and/or status to adjust the actuators' behaviors (e.g., haptic or compression feedback) [14, 17, 25].

Scalable Design, Manufacturing & Sustainability: Scalable design and manufacturing methods with sustainability for soft wearables are critical for widespread adoption. Scaling up soft wearables for everyday, everywhere use brings sustainability concerns to the forefront [2–4, 24].

Deployment for Large-Scale Long-Term Evaluation: Towards everyday adoptions by everyone, large-scale long-term deployment for evaluation is crucial for evaluating intelligent soft wearables in diverse contexts and probing usability.

Soft Wearable Ecosystem: Single-Purpose vs. Generic One-for-All: As the community envisions and invents the future soft wearable ecosystem, one key form factor and interaction consideration on the number of devices in the ecosystem: how many wearables should one own and wear at a time?

¹<https://softwearables.github.io/uist24/>

4 An Estimate of Expected Attendance

We expect to bring together 20-40 attendees interested in intelligent soft wearables and related topics. When we organized the previous workshop at UIST 2024, about half of the attendees brought demos without formal solicitation, so we expect 10-20 demos to be showcased. We will run an open workshop for rich discussions.

5 Workshop Plan

5.1 Pre-Workshop Preparation

Call for Demos & Reports & Papers. Please see Appendix A.

Demo & Reports & Paper Selection and Publication. All submissions will be submitted and reviewed through Precision Conference system (PCS). The review process will be light: each submission will have 2 reviewers who are part of the organizing committee. Authors of the accepted submissions can choose to have their submissions included in the ACM Digital Library.

Call for Participation. We will advertise for open workshop participation not only within the UbiComp/ISWC community but also within other HCI and AI venues via emails and social media. A webpage and repository were created to help promote, organize and document this workshop, along with the previous workshop².

5.2 Day of the Half-Day Workshop

Opening Session (15 minutes). The workshop will commence with a brief opening session to introduce the agenda and objectives for the day.

Keynotes (1 hour). In the morning, we plan to invite 2 keynote speakers in the field to present for 30 minutes. They will talk about insights into the latest advancements and future trends in intelligent soft wearables.

Reports and Paper Presentations (1 hour). Authors for each accepted report/paper will present their work, followed by short Q&A sessions. The exact duration for each report/paper depends on the number of submitted and accepted reports and papers.

Demo Session (1.5 hours). All the demos, both with solicited submissions and informal demos brought by the attendees, will be divided into 2 groups for subgroup demo sessions, aiming to promote collaboration, inspire new ideas, and facilitate networking among researchers, designers, and practitioners in the HCI field. At the end of the demo session, all participants will anonymously vote for a *Best Demo Award*.

Open Discussion on Challenges and Opportunities (0.5 hour). Building on the previous keynotes, presentations, and demos, this session will then focus on exploring opportunities and potential solutions for intelligent soft wearables.

5.3 Post-Workshop Followup

The summary of key takeaways and next steps will be documented on our workshop website to openly share with a broader audience.

6 The Organizers' Backgrounds

Tianhong Catherine Yu is a PhD student at Cornell University, USA. Her current research is on scalable and practical AI-powered smart clothing and textile for human-computer interaction.

²<https://softwearables.github.io/uist24/>

Cedric Honnet is a PhD student at the MIT Media Lab focusing on HCI and advanced manufacturing. With a background in embedded systems, his research explores scalable electronic miniaturization for conformable wearables.

Tingyu Cheng is an Assistant Professor at the University of Notre Dame, USA. His current research is on building scalable and sustainable edge devices for human-computer interaction.

Ryo Takahashi is a Project Assistant Professor at the University of Tokyo, Japan. His research focuses on electronic textiles capable of full-body wireless power and data transmission.

Bo Zhou is Deputy Head of the Embedded Intelligence Department at DFKI. He has a strong interest in smart textile research, especially for AI-driven human activity recognition and interaction.

Cheng Zhang is an Assistant Professor at Cornell University, USA. His research focuses on integrating human-centered AI with advanced sensing technologies to empower everyday wearables in comprehending human behavior in real-world environments, with the goal of better supporting users.

Paul Lukowicz is a Professor of AI at the Technical University of Kaiserslautern in Germany where he is heading the Embedded Intelligence group at DFKI.

Yoshihiro Kawahara is a Professor at the University of Tokyo, advancing IoT sensing with ubiquitous wireless power transfer, ultra-low-powered wearable computing, and instant inkjet circuits.

Josiah Hester is an Associate Professor at Georgia Institute of Technology. He contributes to sustainable computing focusing on battery-free and biodegradable computing systems. He integrates computer engineering, embedded systems, and sustainability to pioneer environmentally responsible ubiquitous computing.

Joseph Paradiso is a Professor at the MIT Media Lab. His research covers wireless sensing systems, wearable and body sensor networks, sensor systems for built and natural environments, energy harvesting, ubiquitous computing, space-based systems, and interactive music/media.

Yiyue Luo is an Assistant Professor at the University of Washington's Department of Electrical & Computer Engineering, USA. Her research interests lie at the intersection of digital fabrication, human-computer/robot interaction, and applied machine learning.

Irmandy Wicaksono is an Assistant Professor at the Division of Industrial Design, National University of Singapore. His research focuses on integrating electronic devices into soft materials for applications spanning physiological sensing, human-computer interaction, interactive arts, architecture, and beyond.

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A Call for Participation

We invite participants who are experienced in and/or interested in intelligent soft wearables, from the design and fabrication of various applications.

We call for three kinds of contributions, including interactive demos, experience/reflection reports, and Work in Progress (WiP)/position papers. Workshop participation is open to everyone, if the venue space allows, without submitting a demo/report/paper.

Each workshop submission will be reviewed by 2 reviewers who are organizers of the workshop. Authors of the accepted submissions can choose to have their submissions included in the ACM Digital Library as supplemental proceedings of the conference.

All accepted submissions will be presented/showcased during the workshop. There will be a Best Demo Award, voted anonymously by workshop attendees.

We call for three kinds of contributions:

- Demos: 2-page submission;
- Experience/Reflection Reports: 4-page submission; and
- WiP/Position Papers: 4-page submission.

Submissions areas of interest include (but are not limited to):

- accessible computational design & digital fabrication toolkits
- scalable manufacturing processes for soft wearables
- sensing, actuation, and/or energy harvesting devices
- sustainable wearables
- applied machine learning for wearables
- multi-function multimodal wearable ecosystems
- evaluating large-scale and long-term deployment
- personalization VS generic one-for-all wearables
- miniaturization for conformability
- wearable technology acceptance